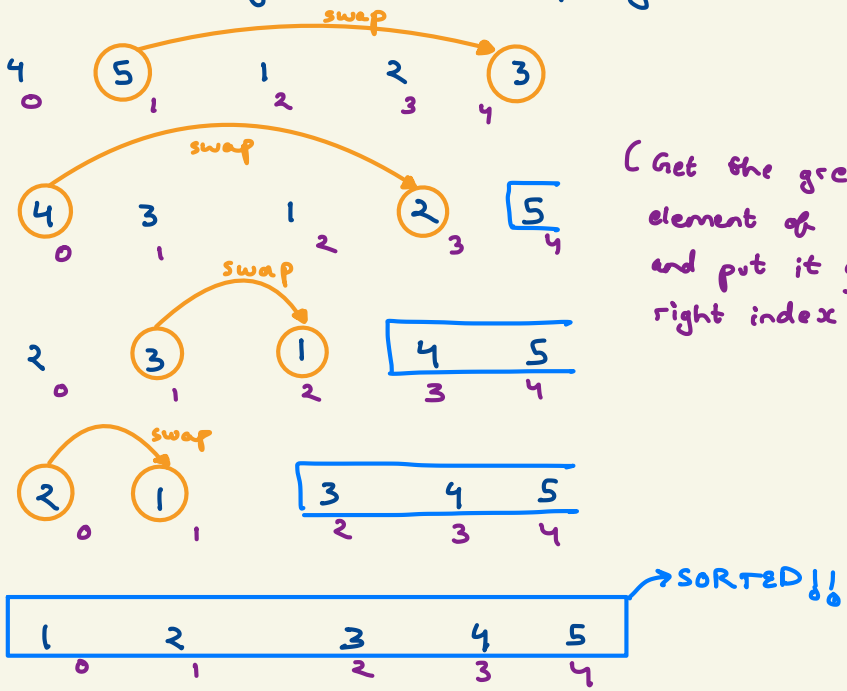
Selection Sort

Selection sort involves selecting an element and putting it on its correct index.

For Eg.)



(Get the greatest element of the array and put it on the right index)

Here, we selected the maximum element, but we can also do it vice-versa.

COMPLEXITY:

Total comparisons $= 0 + 1 + 2 \ldots (n-1)$

$$= \frac{n(n-1)}{2}$$

$$\Rightarrow \frac{n^2 - n}{2} \Rightarrow \boxed{n^2}$$

WORST CASE : $O(n^2)$

BEST CASE : $O(n^2)$

→ It is not a stable sorting algorithm.

→ It performs well on small lists.